

CoRoT-2b

R-0243

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_230717 · created 2026-07-17T05:37:08+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460142.8125 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.224 +/- 0.065
Transit depth (Rp/Rs)^2	5.02 +/- 2.92 [%]
Orbital Inclination (inc)	88.71 +/- 2.06
Airmass coefficient 1 (a1)	1.6 +/- 0.22
Airmass coefficient 2 (a2)	-0.37 +/- 0.73
Scatter in the residuals of the lightcurve fit is	14.08 %
Best Comparison Star	#3 - [626, 147]
Optimal Aperture	4.27
Optimal Annulus	14.82
Transit Duration (day)	0.0992 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.224 $\pm$ 0.065 vs 0.1667 (deviation 0.88 σ)
Duration fit vs published	0.0992 d vs 0.095 d (deviation 0.64 σ)
Mid-time O–C	3.2 min
Depth SNR	3.4
Residual scatter	14.08 %

Light curve

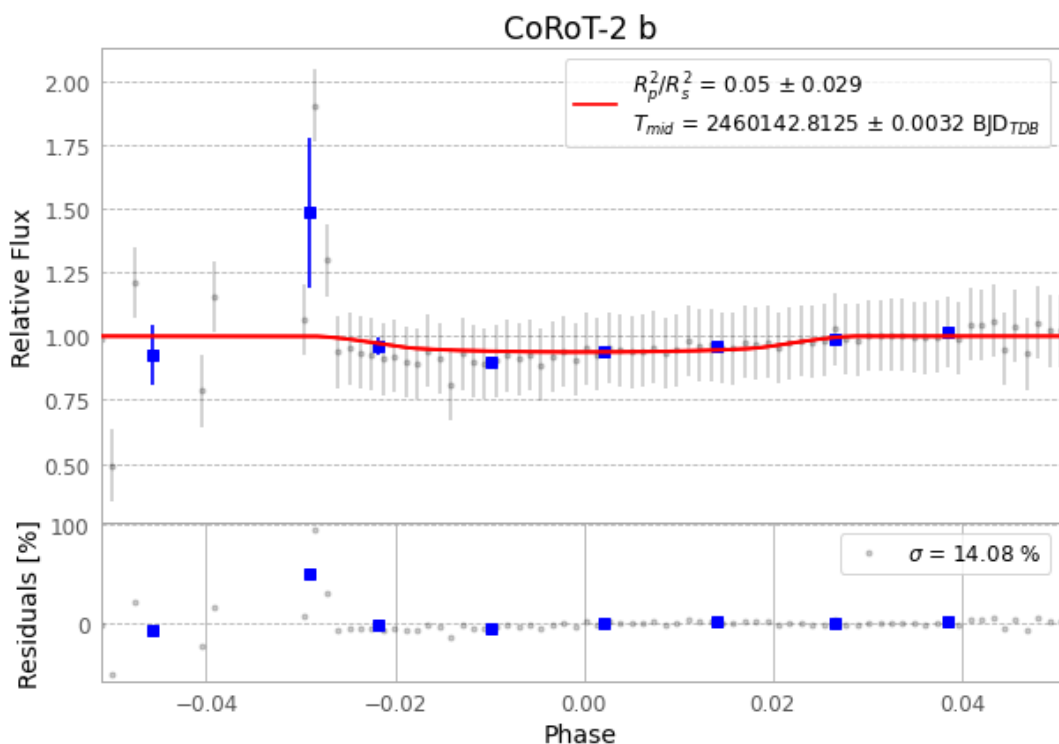


Figure 1 — FinalLightCurve_CoRoT-2 b_2023-07-16.png (SHA-256 4eda0dd054bf917e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [285, 187], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9672337fd8e58814...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

86 FITS frames (56 MB), CoRoT-2230717051215.FITS → CoRoT-2230717092716.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-230717.log (1fab9e62b942...), FinalLightCurve_CoRoT-2 b_2023-07-16.png (4eda0dd054bf...), FinalLightCurve_CoRoT-2 b_2023-07-16.csv (43b677c65f92...)

Compute resources

Wall time 410.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 05:37Z.